

Speech and Gesture Fusion for Multimodal Excel Interaction

Multimodal Interaction

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1 Introduction

This report describes the development of a multimodal system for interacting with Microsoft Excel spreadsheets by combining speech recognition, gesture recognition, and multimodal fusion. This work results from the integration and evolution of two previously developed projects within the *Multimodal Interaction* course — a voice-controlled system and a gesture-controlled system.

Microsoft Excel is an extremely powerful tool, but also a complex one, particularly for users with limited experience using advanced features such as formulas, pivot tables, conditional formatting, or charts. The complexity of the traditional interface based on menus, icons, and technical terminology can significantly increase cognitive load and the risk of errors.

This project proposes an alternative approach based on natural interaction, allowing users to execute complex tasks through voice commands, hand gestures, or a combination of both. Multimodal fusion enables the exploration of key concepts such as redundancy, complementarity, and confirmation of critical actions, making the interaction more robust, safe, and intuitive.

The system was designed with a strong focus on accessibility, efficiency, and error reduction, demonstrating how multimodal interfaces can significantly improve the user experience of traditional applications such as Excel.

2 Persona Definition

To guide the system design and ensure that design decisions are user-centered, a representative persona was defined.

2.1 Persona Profile

- **Name:** Professor Helena Martins
- **Age:** 58 years old
- **Scientific field:** Social Sciences / Education
- **Professional experience:** 30 years of university teaching
- **Digital skills:** Intermediate

The professor regularly uses digital tools such as email, Moodle, and PowerPoint but avoids advanced Excel functionalities due to their complexity.

2.2 Identified Difficulties

The main difficulties faced by the persona include:

- Difficulty using advanced Excel features
- Difficulty calculating averages and creating formulas
- Difficulty applying conditional formatting
- Difficulty generating charts and pivot tables
- Difficulty understanding Excel technical terminology
- Fear of making irreversible errors in the spreadsheet

These difficulties result in a high level of insecurity when working with Excel, leading to slower and more error-prone workflows.

3 Usage Scenario

Professor Helena Martins is preparing her class final grades using Excel. While most grades are straightforward, some students require individual performance analysis across multiple tests.

Working from home, the professor feels more comfortable speaking aloud and using natural gestures, but she requires a reliable system that minimizes errors and does not demand advanced technical knowledge.

Multimodal interaction allows her to:

- Execute complex tasks naturally
- Reduce cognitive load
- Minimize errors
- Work more efficiently and confidently

4 System Architecture

The system follows a modular multimodal architecture, ensuring clear separation of responsibilities and extensibility.

4.1 Input Layer

The input layer includes two main modalities:

- **Speech Modality**
 - Voice capture through a microphone
 - Natural language processing using Rasa 3.1
- **Gesture Modality**
 - Kinect Xbox sensor
 - Visual Gesture Builder

4.2 Fusion Layer

Multimodal fusion is performed by the *MMI Fusion Engine*, which is responsible for:

- Receiving events from multiple modalities
- Combining speech and gesture events
- Implementing redundancy and complementarity
- Generating high-level commands

4.3 Application Layer

The application layer consists of a C# backend that uses *Microsoft Excel Interop* to:

- Execute actions in Excel
- Manipulate cells, columns, and worksheets
- Generate charts and pivot tables
- Provide visual feedback

5 Multimodal Fusion Strategies

Multimodal fusion constitutes the core of the developed system and is responsible for coordinating and integrating events from different interaction modalities — speech and gestures — in a consistent, robust, and safe manner. This approach allows the system to leverage the strengths of each modality while reducing ambiguities, minimizing errors, and increasing user confidence during Excel interaction.

The system explicitly implements three fundamental multimodal interaction strategies: *single-modality interaction*, *redundant interaction*, and *complementary interaction*. These strategies were defined based on established principles in multimodal interaction research and were carefully applied to the Excel functionalities supported by the system.

5.1 Single-Modality Interaction

In single-modality interaction, a given action is executed exclusively through a single modality, without requiring confirmation or support from another modality. This strategy is primarily used for simple, frequent, and low-risk actions, where execution speed is prioritized and the probability of error is low.

In the developed system, the gesture modality is particularly suitable for continuous navigation tasks, while the speech modality is more appropriate for commands that do not require spatial references.

Examples of single-modality interactions include:

- Zooming in and out using gestures

- Vertical and horizontal navigation within the spreadsheet using gestures
- Generating class performance charts using voice commands
- Generating individual student charts using voice commands
- Generating test question charts
- Deleting all charts from the worksheet

Using single-modality interaction reduces system complexity and ensures smooth and immediate interaction for simple actions, without introducing unnecessary confirmation delays.

5.2 Redundant Interaction

Redundant interaction occurs when the same action can be executed using more than one modality. This strategy increases system flexibility and adapts to user preferences or contextual constraints of the interaction environment.

In the proposed system, several Excel actions can be executed using either voice commands or gestures, allowing users to choose the modality that feels most comfortable at any given moment.

The main actions supported through redundant interaction include:

- Calculating averages
- Inserting columns
- Highlighting approved students
- Highlighting failed students

For example, users may calculate the average of a selected range using a voice command or alternatively by performing a predefined gesture. This redundancy improves accessibility and reduces error likelihood by allowing users to rely on their preferred modality.

5.3 Complementary Interaction

Complementary interaction represents the most advanced level of multimodal fusion implemented in the system. In this type of interaction, action execution requires the mandatory combination of two or more modalities, with each modality providing part of the information needed to perform the command correctly.

This strategy is mainly applied to complex, sensitive, or potentially destructive actions, where precision, correct context, and explicit user confirmation are essential.

In the developed system, complementary interaction is applied in the following scenarios:

- Selecting cells or ranges using the *Hand Grab* gesture, followed by a voice command to calculate averages or generate charts

- Selecting students through gestures, followed by voice commands to apply conditional formatting
- Inserting columns using voice commands, followed by dependent actions on the new structure
- Executing the *Undo* action using a gesture, followed by verbal confirmation
- Closing Excel using a gesture, followed by voice confirmation

This mechanism significantly reduces the risk of critical errors, such as accidentally closing the application or applying operations to incorrect ranges. Requiring multimodal confirmation forces users to explicitly validate their intentions, increasing system safety and trust.

5.4 Fusion Engine Behaviour

Coordination between modalities is managed by the *MMI Fusion Engine*, which receives low-level events from speech and gesture modalities and transforms them into high-level commands understood by the Excel backend.

The Fusion Engine is responsible for:

- Synchronizing multimodal events in time
- Determining whether an event should be treated as single, redundant, or complementary
- Resolving ambiguities between competing commands
- Implementing confirmation mechanisms for critical actions

This approach ensures that the system behaves consistently and predictably, even in scenarios where multiple events are detected within a short time window.

5.5 Impact on Usability and Error Reduction

The structured application of multimodal fusion strategies has a direct impact on system usability. Combining modalities enables the system to:

- Reduce user cognitive load
- Decrease the probability of errors
- Increase task execution efficiency
- Improve accessibility for users with different skill levels

By effectively leveraging redundancy and complementarity, the system demonstrates how multimodal interaction can overcome the limitations of traditional mouse-and-keyboard interfaces, providing a more natural, safe, and efficient user experience.

6 Challenges

Developing the multimodal system involved several technical and interaction challenges arising from the integration of multiple modalities and components, as well as the need to ensure a fluid, reliable, and safe user experience.

6.1 Multimodal Recognition Accuracy

Ensuring accurate multimodal recognition was one of the main challenges of the project. Both speech and gesture recognition are subject to uncertainty caused by speech variability, background noise, lighting conditions, and gesture execution differences. Combining these modalities required robust fusion mechanisms capable of handling ambiguous events and avoiding false positives without compromising system responsiveness.

6.2 User Variability Across Modalities

User variability in speech patterns and body movements made it difficult to define universal commands that work consistently for all users. To mitigate this issue, gestures were designed to be clearly distinguishable, and confidence thresholds were carefully tuned to balance accuracy and usability.

6.3 Real-Time Responsiveness and Latency

Real-time latency was a critical factor, as perceptible delays between user interaction and system response negatively affect user trust. Optimizing communication between recognition modules, the fusion engine, and the Excel backend was necessary to ensure near-instant visual feedback.

6.4 Integration of Multiple System Components

Integrating multiple independent components, including sensors, recognition modules, fusion engines, and the Excel backend, required a well-defined modular architecture. Ensuring stable communication and temporal synchronization between components was essential to avoid inconsistent system states.

6.5 Coordination Between Modalities

Effectively coordinating multiple modalities was particularly challenging when determining when to use single-modality, redundant, or complementary interactions. Critical actions required explicit multimodal confirmation to prevent accidental execution while maintaining a smooth interaction flow.

6.6 Balancing Safety and Usability

Finally, balancing safety and usability was a key challenge. While confirmation mechanisms increase system reliability, excessive confirmations can introduce friction. The system applies confirmations only to sensitive actions, ensuring safety without sacrificing efficiency.

7 Conclusion

This project demonstrates that multimodal interaction can transform traditional applications such as Excel into more accessible, efficient, and intuitive tools. The combination of speech and gesture interaction reduces cognitive load, minimizes errors, and significantly improves the overall user experience.

Future work includes expanding the multimodal command set, improving contextual fusion mechanisms, and supporting additional usage scenarios.

8 References

- Microsoft Excel Interop Documentation
- Rasa NLU Documentation
- Kinect and Visual Gesture Builder Documentation
- Lecture Materials – Multimodal Interaction